

ZAHRA JENABZADEH

BSc in Computer Engineering

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Higher Education

2018 – 2023 | **B.Sc. in Computer Engineering**
Shiraz University, Shiraz, Iran
GPA of last 4 semesters: **18.3/20 (3.87/4)**
- **Dean's list last 5 semesters**
CGPA: 16.84/20 (3.46/4)
- **Among the top 20% of the class**
GRE: **316** (Q: 165/170, V:151/170, AW:3.5/6)

Publications

- Javad Dogani¹, **Zahra Jenabzadeh**², Mojtaba Roshani³, Farshad Khunjush⁴, “SAFE-PFL: Efficient and Secure Personalized Federated Learning”, (weak acceptance in ACM SIGMETRICS)
- **Zahra Jenabzadeh**¹, Hooman Tahayori², “An Unsupervised Fuzzy Rule-Based System for Emotion Detection”, National Conference of Artificial Intelligence and Software Engineering, (Aisoft23), Shiraz, Iran, November 2023, civilica.com/doc/1912891/
- Dr Samad Doostdar¹, Dr Hamideh Pourtaherian², Seyed Mohammadhessam Mohammadi³, Anahita Khanmohammadi⁴, and **Zahra Jenabzadeh**⁵, “A Muti-Agent AI Framework for Scalable and Context-Aware Job Recruitment Process”, ADCAIJ, August 2025.
- Dr Samad Doostdar¹, Dr Hamideh Pourtaherian², Seyed Mohammadhessam Mohammadi³, Anahita Khanmohammadi⁴, and **Zahra Jenabzadeh**⁵, “Bridging the gap: An AI-Driven Framework to Address Recruitment Challenges in Canada”, ADCAIJ, August 2025.

Research Experiences

May
2026 -
present | **Research Assistant**
Shiraz University, Shiraz, Iran (remote)
Under the supervision of Dr. Moosavi
Research Area: Geometric Deep Learning, Computer Vision, Transformer-based Learning, Pattern recognition
- Developed a transformer-based framework to convert rooftop perspective images into true ortho representations by detecting rooftop polygons and mapping polygon vertices between perspective and ortho coordinate spaces using local and global graph transformers.

Dec.
2026 -
present | **Research Assistant**
Shiraz University, Shiraz, Iran (remote)
Under the supervision of Dr. Moosavi
Research Area: Teeth Segmentation, Image Processing, Transformer Models, Finetuning, Image Generation
- Segmenting teeth using state-of-the-art transformer models. Evaluating results and enhancing accuracy of models using data augmentation. Furthermore, generating teeth images with the help of diffusion models.

May
2024 -
Oct. 2025 | **Research Assistant**
IMDEA Networks Institute, Madrid, Spain (remote)
Under the supervision of Dr. Dogani
Research Area: Distributed Learning, Fine-tuning Pre-trained LLMs, Federated Learning, Optimization, NLP
- Fine-tuning LLMs using PEFT methods such as **LoRA** in distributed settings to reduce computational and communication overhead.

Research Assistant
IMDEA Networks Institute, Madrid, Spain (remote)
Under the supervision of Dr. Khunjush & Dr. Dogani
Research Area: Edge Computing, Privacy Preservation, Personalized Federated Learning, Distributed Learning, Secure Clustering, Selective Parameter Encryption, Decentralized Learning

- A novel scalable framework that optimizes PFL by enhancing privacy, encrypting just 10% of the model without sacrificing model performance. Additionally, it effectively guards against reconstruction attacks with minimal computation overheads of 12.68%
- **Privacy preservation techniques:** multi-key homomorphic encryption, SGD Differential Privacy, Secure Clustering
- **Models:** CNN, ResNet-18, MobileNet-V2, ResNet-50, VGG-16, ViT-Base, ViTMAE
- **Data distribution techniques:** Imbalanced label distribution skew, Dirichlet Distribution, IID
- **GitHub Link:** [SAFE-PFL](#)

Apr. 2025 **AI Researcher**

- Aug. *Canada Prime Jobs Inc. Vancouver, British Columbia, Canada (remote)*

Research Area: Applied AI, Fine-tuning Pre-trained LLMs, NLP, Prompt Engineering, HR Management

- A novel multi-agent system, using Google's Gemini models that extract information from job applicants' resumes and find the best match for the recruitment process.

Mar. 2023 **Research Assistant**

- Oct. *Shiraz University, Shiraz, Iran (on-site)*

Under the supervision of Dr. Tahayori

Research Area: XAI, NLP, Fuzzy Logic, FRBS, ML, Deep Learning, Sentiment Analysis, Text Classification

- An unsupervised fuzzy rule-based system to classify tweets into 3 categories. Results were compared with state-of-the-art/data-driven methods and 3 fuzzy rule-based systems, showing that our proposed fuzzy system improved accuracy by approximately 10% compared to the aforementioned fuzzy systems.
- I explored sentiment extraction from tweets using fuzzy logic. Unlike data-driven models such as **LSTM** and **CNN**, my approach utilizes a rule-based system with nine rules and trapezoidal membership functions. I comprehensively compared my model against other methods, including **Naïve Bayes** and **SVM** using **TF-IDF Vectorizer**, across five datasets from Kaggle and GitHub. The results demonstrated superior performance in three out of the five datasets. Furthermore, I incorporated **Reinforcement Learning** with fuzzy logic to optimize lexicon selection for Twitter sentiment analysis, finding that Vader and AFINN lexicons yielded the best results.

Teaching Assistantships

May 2022 **Design and Principles of Algorithms**

Dr. Zohreh Azimifar | Shiraz University, Shiraz, Iran

- Designing programming assignments, grading, holding TA classes

May 2019 **Design and Principles of Algorithms**

Dr. Koorush Ziarati | Shiraz University, Shiraz, Iran

- Designing programming assignments/projects, grading, holding TA classes

May 2019 **Data Structures and Algorithms**

Dr. Hossein Esfandiari | Shiraz University, Shiraz, Iran

- Designing programming assignments, grading, holding TA classes

Work Experiences

Apr. 2025 – **ML Engineer**

Aug. *Canada Prime Jobs Inc. Vancouver, British Columbia, Canada (remote)*

- Building and training AI agents. Finetuning the AI models using prompt engineering and RAG system.

Mar. 2023 – **Mathematics' Tutor**

Sep. *Alvand Institute, Shiraz, Iran*

- Educating high school students in discreet mathematics, geometry, calculus, statistics to prepare themselves for the university admission exam.

IT Specialist

- Helped with UI/UX design and implementation of the institution's website.

Jul. 2022 – **Web Development Intern**

Sep. *Asre Andishe, Shiraz, Iran*

- Designing a web application that reads an XML file and displays the contents on a map. Other features of the website include: Filtering locations based on user inputs

using WFS and jQuery and displaying population layers using WMS.

Skills

PROGRAMMING: Python, Java, C, JavaScript
LIBRARIES & Tools: TensorFlow, Scikit-learn, Keras, PyTorch, Langchain, Flower, OpenLayers
WEB PROGRAMMING: HTML, CSS, Django, Flutter, Vue, UI/UX, Docker
CONCURRENT PROGRAMMING: Python Multithreading, Python Multiprocessing
DATA MINING: NumPy, Pandas, R (Programming Language)
PROJECT MANAGEMENT: Git, Agile Software Development (SCRUM)
RESEARCH: Scientific Writing with LaTeX, Statistical Analysis
DIGITAL PAINTING: Illustrator, Procreate, Blender

Languages

- English: IELTS 8 (Listening: 8.5 Reading: 7.5 **Writing: 7.5 Speaking: 8.5**)
- Germany: A1
- Persian: Native

Selected Projects

- Aug. 2025 **BEHRAVAN Multi-agent AI Assistant**
Language & Tools: Python, Langchain, streamlit
Objective: Training a Multi-agent AI assistant to detect users' mental health problems and refer them to a sub-agent to inquire further questions and give recommendations. Providing data for the models was established through a RAG system.
- Jan. 2025 **GPT-2 Decipher**
Language & Tools: Python, PyTorch, Numpy
Objective: Implementing a model inference function for a fine-tuned GPT-2 model to decipher encrypted text.
- Oct. 2023 **Spotify Playlist**
Language & Tools: Python, PyTorch, Numpy
Objective: With the help of data analysis and ML, design a model to estimate a song's total plays and streams, and a model to predict whether a song is a single or an album.
- Jul. 2022 **GIS Web Development (Internship Project)**
Language & Tools: HTML, CSS, JavaScript, Open Layers, GEO Server
Objective: A website to display contents on the map from an XML file uploaded on the geo server.
- Jul. 2022 **CISCO Network Management**
Course: Computer Networks
Language & Tools: CISCO Packet Tracer
Objective: Reconfigure the root router, set/recover password for a router, and update the iso image on a flash.
- Jul. 2021 **Train SE Web Development**
Course: System Analysis and Design
Language & Tools: Vue, Bootstrap, Django
Objective: A website for students to learn about System Analysis and Design.
- Jan. 2021 **Compiler Implementation**
Course: Compiler Design
Language & Tools: Python
Objective: Design and implementation of a compiler with given rules.
- Jul. 2020 **LinkedIn**
Course: Database
Language & Tools: SQL, Flutter
Objective: A website for developers to upload their works and skills.

Certificates

Oct. 2024	Generative AI with Large Language Models Score 100% / COURSERA / Deeplearning.AI
Nov. 2023	Introduction to Probability and Data with R Score 100% / COURSERA / Duke University
Oct. 2023	Mathematics for Machine Learning: Linear Algebra Score 100% / COURSERA / Imperial College London
Oct. 2023	Supervised Machine Learning: Regression and Classification Score 100% / COURSERA / Stanford University
Sep. 2023	Machine Learning Workshop Score 100% / CSESA / Scientific Association of Computer Engineering of Shiraz University
Sep. 2023	Natural Language Processing with Sequence Models Score 100% / COURSERA / Deeplearning.AI
Jul. 2023	Natural Language Processing with Classification and Space Vectors Score 100% / COURSERA / Deeplearning.AI

Selected Courses (GPA: 4/4)

Computer Networks: 20/20	Algorithm Design and Analysis: 18.75/20
System Design and Analysis: 19.7/20	GPU: 18.5/20
Design of Embedded Systems: 19.5/20	Graph Theory and Algorithms: 17.97/20
Artificial Intelligence: 19/20	Green Computing: 17.3/20

Volunteer Experiences

Nov. 2023	IEEE Women in Engineering - Assistant of the president (Dr. Zohreh Azimifar) - Designing a poster for the event and setting a virtual environment for online viewers.
Apr. 2022	SPARK competition (German competition held by Sharif and Shiraz University) - Object Detection for Blind People. - Language: Python, OpenCV, Arduino
Oct. 2020	Hour of Code (HOC) - An event organized by the ACM association for high school students.

References

Will Be Provided Upon Request.